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POLICY BRIEF

GENERAL-PURPOSE GENAI CHATBOTS: PRESERVING THE PUBLIC INTEREST

Author:

Paula Gori

Florence School of Transnational Governance

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EXECUTIVE SUMMARY

Generative Artificial Intelligence (GenAI) tools – in particular, general-purpose GenAI chatbots – are now widely used by society. The training methodology and the quality of the data on which they are trained impacts their outputs. These tools are trained on large sets of mainly public data, and their outputs (i.e., responses provided to the user) depend on the prediction of the next word based on the statistically most-used word in a given context (they do not possess knowledge or reasoning capacities *per se*). As such, their responses may be biased, misleading, or inaccurate (a phenomenon that is often referred to as 'hallucinations'). These tools, which have high energy and carbon impacts, can also negatively impact our neural and cognitive capacities. Policy responses should support a balanced use of general-purpose GenAI chatbots, integrate responses across sectors (including those outside the technology sphere), support research, and ensure that tech companies act in the public interest.

Author:

Paula Gori | Programme Coordinator, Florence School of Transnational Governance, EUI

Views expressed in this publication reflect the opinion of individual authors and not those of the European University Institute.

1. INTRODUCTION[1]

A recent study by MIT researchers looked at the cognitive consequences of using Large Language Models (LLMs) for academic essay writing. In particular, researchers assigned participants to three groups – the LLM group, the search engine group, and the brain-only group – and asked each participant to use a designated tool (or no tools, in the case of the brain-only group) to write an essay. Researchers concluded that, while LLMs offer immediate convenience, they also generate potential cognitive costs. Indeed, LLM users consistently underperformed members of the other two groups on neural, linguistic, and behavioral measures. Among the costs of using LLMs, researchers observed that LLM users were less inclined to critically assess the outputs of the LLM. This study, along with others like it, heralds the advent of a proliferation of research on this topic.

This study was published a few weeks after Policy Leader Fellows at the EUI Florence School of Transnational Governance organised a workshop on 'AI and the war against democracy. Risk, resistance and responsibility,' also aimed at promoting discussion of the pros and cons of generative AI technology and unpacking the reasons behind the widespread appeal of conversational chatbots like ChatGPT. There are, of course, many reasons for ChatGPT's popularity; one such reason that emerged during workshop discussions was the imperative, starting at a very early age, for individuals to come up with exact answers that are judged (if not rated) by educational and professional institutions. Be it in school, among friends, during a job interview or at the workplace – in countless real-life scenarios, people are expected to provide exact answers and may be penalised if they fail to do so. In this constant 'race' in which the front-runner is the one that comes up with the requested number, name, or response, LLMs appear to be a sort of relief. To those desperately seeking a precise answer, these tools appear to be a sort of panacea or magic wand, providing what seems to be a definitive answer within the blink of an eye (an immediate reward). Not by chance, in March 2025, ChatGPT became the most downloaded app globally (excluding games).

2. PANDORA'S BOX

During the conversation, there was agreement that the relief that LLMs seem to offer is, in large part, an illusion. The validity of responses provided by ChatGPT, Gemini, Claude, and other such LLMs are, simply put, dependent on how and with which data the model was trained. Hence, while answers may be accurate, they may also be biased, misleading, or inaccurate. Answers provided by these tools are 'human-like' in the sense that they are grammatically correct and drafted in a form that appears professional and reliable. This is reassuring and lowers the defences and critical thinking reflexes of LLM users, giving them the feeling of saving time and avoiding the negative consequences that may stem from an incorrect answer. The risk of ending up in an automation bias situation is also present, meaning a tendency to trust more in and rely more on GenAI chatbot responses than one's own knowledge and intuition. Use of these tools can diminish users' doubts and scepticism and dampen the instinct to counterargue, all of which are important aspects of critical thought. Notably, a seemingly 'opposite' phenomenon is also present: as generative AI use proliferates, we may find ourselves in a constant 'doubt bubble,' regularly asking ourselves if the online content in front of us is AI-generated or not. Along with a general sense of heightened mistrust, generative AI's spread also opens the doors for the 'liar's dividend' – in other words, people may reap benefits from spreading or legitimising disinformation.

Investing in research to understand these phenomena and their impacts will be a key part of mitigating their harmful effects.

3. A TROJAN HORSE?

Text-based GenAI all-purpose conversational chatbots that are typically used by the public (such as ChatGPT, Gemini, and Claude) are

1 Paula Gori is the Secretary-General and Coordinator of the European Digital Media Observatory (EDMO), the EU's largest interdisciplinary network to counter disinformation. EDMO is based at the EUI's Florence School of Transnational Governance.

largely trained on publicly available online data. Concerning the possibility of using copyrighted content to train such models, high-level debates are currently taking place on how to ensure fair remuneration and whether the traditional concept of copyright will still be appropriate for these new contexts.

The question of the quality and reliability of these LLMs is a pressing one. Public data is also a limited resource, and AI-generated content is increasingly entering the basket of publicly available online data used to further train GenAI. This cyclical use of AI-generated content, according to research, may be problematic. OpenAI, for example, states that ChatGPT models are developed "using three primary sources of information: (1) information that is publicly available on the internet, (2) information that we partner with third parties to access, and (3) information that our users, human trainers, and researchers provide or generate."

It may be worth recalling that these LLMs generate responses by predicting the next word based on the statistically most-used word in a given context; they do not possess knowledge per se. This calls the appropriateness of the moniker 'artificial intelligence' into question (for a fuller discussion of this debate, see the following reading by Luciano Floridi).

GenAI can be used for malign purposes, such as producing and distributing disinformation content. LLMs are often also biased – for example, against women. Training all-purpose LLMs with quality data is key to avoiding production of misleading or inaccurate content. Users also hold some responsibility in their consumption and use of online content. For example, the risk of snowball effects resulting from the use of GenAI-produced misinformation or gender-biased content to further train LLMs – which, in turn, run the risk of generating similarly misleading or prejudiced content – is easy to see.

To put part of GenAI-related controversies in perspective, it is worth recalling that the news industry has seen a drastic change in the last decades. The growth of online publications ensured access to a much larger volume of information compared to traditional media sources, including content produced by a wider range of actors. This, however, came at a price. The business model of traditional media was heavily impacted, with resources for quality control and protection of journalists under threat and advertisers increasingly moving to the online market. Sensationalist content ensured virality and monetisation to the detriment of factuality and accuracy, copyright battles raged, and the public's media literacy skills struggled to catch up with the pace of technological change. With the birth of the attention economy, the meaning and value of news changed radically. This stage is now populated by many actors, including bad ones who misuse the system for malign purposes. In the present moment, GenAI tools like ChatGPT and Gemini are entering the scene both as content producers and as content users (for model training). As a result, it is necessary to question the quality of the content that these GenAI tools produce. As a matter of example, experts from American Sunlight highlighted that the Russian Pravda Network (a series of pro-Russian disinformation-spreading websites) is used as a tool for training LLMs. In other words, these websites, which may sometimes appear innocuous at first, play the role of generating online disinformation content to be captured as data for LLM training, which helps to produce biased responses. The consequence of this cascade is quite self-evident.

4. GOOD DATA FOR GOOD OUTPUT

Public and private institutions are developing their own LLMs trained with these institutions' own data, though operating and maintenance costs are steep. Very recently, the Italian Parliament presented NORMA, MSE, and DEPUCHAT. These three AI-powered chatbots are aimed, respectively, at producing reports, graphs, and tables of legislative processes, as well as identifying amendments, legislative texts, and other documents to support members of Parliament and their teams; accompanying members of Parliament and their teams during the preparation of amendments; and providing citizens with information on activities carried out by specific members of Parliament on given topics and areas in order to foster public transparency and accountability. These

are examples of GenAI easing the work of public institutions and supporting the work of professionals, made possible by training the models with reliable internal data.

Moving to the media sector, an example worth mentioning is the pilot test of an LLM recently launched by the BBC. The latter is trained with BBC content only and provides short summaries of BBC articles (reviewed by journalists). In this case, training data and output will very probably be more reliable than less selectively trained LLMs, as all training data follow the transparent quality guidelines of the BBC. These types of LLMs, trained with sector-specific quality data and supporting specific purposes, may represent an important market opportunity for the EU, even allowing it to take the lead in this form of AI development.

5. TECHNOLOGY TO FIX PROBLEMS, NOT TO CREATE THEM

Debates around GenAI, and AI in general, inevitably raise the question of where we, as human beings, are headed. In this constant race to save time, often at the expense of not just quality but also our neurological functioning, we must confront questions about whether these tools are needed at their current intensity and pace – and, if so, for what purposes and where to set limits.

As noted by Vladimír Šucha in a recent STG Policy Brief, generative AI use may be associated with negative cognitive impacts (such as attention erosion and emotional dependency), erosion of human capabilities (also called de-skilling), and other hidden costs, such as loneliness and mental health issues.

Humans, the creators of generative AI, must take an active role in governing its use; indeed, AI bills are being adopted all around the world. Ensuring that different forms of expertise are included in these governance exercises may help us prevent damage before it is too late. Tech and AI governance are less about markets and more about human rights, health, climate impact, safety, sociology, and ethics, among other concerns. Particularly given that certain features of GenAI chatbots may have a negative impact on neurological functioning and even cause addiction, policy approaches to tech regulation should adopt a multidisciplinary approach, bringing in aspects of health, environmental, and educational policies. This should be done in a smart way, leveraging behavioural insights rather than imposing changes or mandates. Technological progress should be in the best interests of citizens. We should be the ones dictating the development agendas of tech companies, not the other way around.

6. IF WE CAN DO IT WITH CARS, WE CAN DO IT WITH GENAI

The invention of cars allowed people to go from point A to point B without expending much bodily energy. However, it created new governance needs, such as building proper streets, defining traffic laws and security norms, setting speed limits, and creating a licensing system. Despite being a way to move that does not require much human bodily energy consumption, it soon became evident that this does not mean that going from one point to another by car is always the best option. For example, if your destination is nearby and you are in good health, it is often preferable to walk: to stay healthy, to avoid the need to look for a parking space, to have a lower carbon impact, and so on. On other occasions, even when your destination is far away, you might opt for public transport, as it is more climate-friendly and allows you to avoid traffic, chat with people, or read a book while traveling.

A similar approach could be applied to general-purpose GenAI chatbots. We should use these tools when doing so really makes our lives easier and approach AI use with a long-term sustainability perspective, rather than simply using it out of laziness or to generate 'AI slop' (low-quality content); furthermore, we should only use these tools if we have the skills to deal with the output we get. General-purpose GenAI chatbots can generate misinformation and carry risks of malign political or economic manipulation and the overall carry an error margin; they also have rather high energy and carbon impacts. This stance is not about being against technological development; on the contrary, it is about reminding us that

technology should work in our best interest as humans. Tech should help to fix problems, not create them.

7. POLICY RECOMMENDATIONS

1. **Online and offline should be considered as one world and not as two separate ones:** We cannot continue thinking that online and offline are two separate worlds. On the contrary, they often overlap and interact; what happens online impacts the offline world and vice versa (e.g., hate speech online turning into hate crimes offline). As a result, talking only about digital regulation misses the point. We have regulation regimes in different sectors offline, and this has to be reflected online as well.

2. **Tech policy should be integrated with other sectors:** In the EU, the Digital Services Act, the AI Act, and the General-Purpose AI Code of Practice, among others, are fundamental tools for risk assessment and risk management. However, a vertical policy approach is a blind one. It should, on the contrary, be horizontal and look at other sectors, such as health and climate. Focusing on health, the European Parliament's 2023 call for EU rules to address digital addiction is a first step. In the recent 2025 State of the Union speech, European Commission President Ursula von der Leyen said that she is "appalled by the disinformation that threatens global progress on everything from measles to polio." Policy actions should look health damage caused both by online social media and GenAI platforms' functioning and by the content shared and produced on these platforms. The same is true for the climate impacts of GenAI. Online traffic, and particularly GenAI tools, consume massive amounts of energy and, in parallel, distribute and generate climate disinformation, causing negative downstream impacts on climate action. These are just two examples showing the need to look at tech regulation horizontally, addressing its crossovers to other sectors. As a consequence, multi-stakeholder and multi-disciplinary fora for policymaking should be supported and promoted.

3. **The tech market should be driven by our society, not by private companies:** Market dynamics can be varied; put in simple terms, the needs and tastes of consumers may sometimes create a market space for providers, while at other times commercial operators may create the market space by alerting consumers to needs of which they were previously unaware. Considering the huge impact that technology has on human rights and on all aspects of our lives, from democratic processes to content access to daily activities and health, the tech market should be driven by citizens' needs and the public interests, and not the other way around. Those needs should be matched against the negative externalities of a given product or service, such as potential addiction or cognitive impacts. Tech companies should work in close collaboration with practitioners with expertise in human rights, environmental policy, public health, anthropology, sociology, and law at the brainstorming stage of any new development, helping to ensure that such development goes in the direction of the public interest and citizens' wellbeing.

4. **Set international standards for public wellbeing impact assessments in the short, medium, and long term:** In line with international human rights standards-setting processes, indicators should be developed and transparently implemented to assess the impact that new tech developments have on the overall wellbeing of citizens. Such standards, as well as impact assessments, should be independent and capable of identifying and weigh all positive and negative impacts. The results of such exercises should inform decisions about whether or not to proceed with given developments.

5. **Education and awareness-building policies should be implemented as lifelong learning programmes:** Education policies typically fall under the sovereign jurisdiction of states, and the EU is no exception. National policymakers should be invited to design and implement education and awareness-building programmes

around GenAI, which should meet a number of requirements. Namely, they should be updated very frequently to account for the quick pace of tech development, be implemented as lifelong learning programmes (available not just to young people but to all generations, and empower citizens with necessary critical skills without dampening trust in institutions. Regional and international organisations should support this work by establishing principles and guidelines. Awareness-building programmes could look to other sectors for messaging guidance and inspiration. For example, food-related metaphors could be used to communicate the risks and benefits of GenAI chatbot use. While a moderate amount of time spent using social media and general-purpose GenAI chatbots (akin to enjoying a small amount of dark chocolate) might have more pros than cons and not pose much of a risk to cognitive critical skills, spending too much time on those tools (like eating an entire large bar of chocolate every day) could have more negative consequences. In addition to fostering critical thinking skills, educational programmes could also seek to help citizens identify and engage in offline activities that support emotional balance, such as sports, music, painting, and meditation.

6. **Financially support research in different fields:** The impact of current and new AI developments on human beings in the short, medium, and long term should be assessed, and findings should be integrated into market and policy decisions.

7. **Financially support executive education programmes in psychology and related fields:** Psychologists should receive additional education that allows them to understand and address the impact of social media and GenAI use on patients. Their professional capacities need to be updated, and research projects on this topic must be supported.

8. CONCLUSION

General-purpose AI tools provide immediate answers whose quality should always be questioned (even when responses ultimately end up being factual and correct). They are attractive in terms of time efficiency, but less so in terms of quality and, according to recent research, in terms of neural impact. Technological development is not new, nor is it something to be stopped entirely. However, given the growing role of generative AI in the infrastructure of daily life, its development should be grounded in the public interest and in respect for fundamental rights. Research should be funded to understand these tools' impacts (both positive and negative), and access to lifelong learning tools to deal with new technological developments should be guaranteed to all citizens.

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Florence School of Transnational Governance
European University Institute
Via Camillo Cavour, 65a, 50129 Firenze (FI), Italy
Tel. +39 055 4685 545
Email: stg.publications@eui.eu

www.eui.eu/stg

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